

Lab: two servers, one plug

about 45 minutes

You're going to connect Claude Code to two live systems and then ask it questions neither one could answer alone.

The classroom Wi-Fi is a Mist network. A few Hamina Clips around the room are running as Orb sensors, measuring what the Wi-Fi feels like from a client. Mist knows the network's side of the story. Orb knows the client's. Both have an MCP server.

You'll be handed a key card with the keys you need. They're read-only, were made for today, and get revoked tonight. They go in a file, not in the chat. This lab stays on your laptop; nothing gets pushed.

PART 1: CONNECT BOTH SERVERS

A new session. In the desktop app, open the Code screen and start a new session in a new, empty folder: ask Claude to make one called `mcp-lab` on your Desktop if you need to, then start a session in it.

Point Claude at the docs. Have it make you a credentials file to fill in:

```
Read these two pages and work out what each MCP server needs to connect:
```

```
Juniper Mist: https://www.juniper.net/documentation/us/en/software/mist/mist-aiops/shared-content/topics/concept/juniper-mist-mcp-claude.html
```

```
Orb Cloud: https://orb.net/docs/integrations/mcp
```

```
Create a file called .env in this folder with one line per value each server needs, left blank, with a comment above each line saying what it is and where I'd get it. Don't add the servers yet.
```

Fill in the keys. When it's done, a panel in the chat lists the files it made. Click `.env` there and it opens in an editor. Fill in the blanks from your key card, one value per line, and save. The comments Claude wrote say which value goes where.

Connect. Have Claude add the servers, reading the keys from the file:

```
Add both MCP servers to Claude Code at user scope, using the values in .env. For Orb use the API key from the file, not OAuth. Read the values from the file; don't paste the keys into this chat. When both are added, tell me the two commands you ran.
```

Read the two commands it ran. Each one should be `claude mcp add` with a URL and one or more `--header` flags.

A new session. A running session doesn't see servers added underneath it. Start a new session in the same folder.

Check they're there. Type `/mcp` and press return. A panel lists the servers; `mist` and `orb` should both be there with a tick. Close it.

Read the menu. The panel doesn't show what each server offers, so ask:

```
How many tools does each MCP server give you? List the Mist tools, with what each one does.
```

Note the two counts. The list is the menu from the lecture: the only thing Claude knows about the service.

PART 2: ASK THE NETWORK

Ask each of these and watch the tool calls go past before the answer arrives.

Mist only.

```
List the access points in this org, with their site, model and whether they're connected.
```

Mist only.

```
How many clients are connected right now, and which band, channel, and SSID is each one on?
```

Orb only.

```
Which Orbs are online, and what is the current Orb Score for each?
```

Both. This is the one the lab exists for:

```
Take the Orb with the lowest score right now. Using both Mist and Orb, tell me whether the problem looks like the Wi-Fi link (the AP it's on, that AP's client load, channel and retries) or something upstream of the AP (its responsiveness to the internet). Show me the evidence from each system before you give a verdict.
```

Your own. Ask one question of your own that needs both systems to answer. Something you'd want to know about this room's Wi-Fi, or about your own network if these were your servers. Watch which tools it reaches for.

STRETCH: MAKE IT A SKILL

If you're done early, pick the question you found most useful and turn it into a skill, the way you did in the skills lab:

```
Make a skill called network-check that runs the comparison from my last question against whatever Orb has the lowest score, and reports the evidence from both systems before the verdict.
```

Then start a new session and invoke it.

YOU'RE DONE WHEN

- `/mcp` shows both servers connected.
- You have one answer on screen that used tools from both servers.
- You can say, in one sentence, what Claude read to decide which tool to call.